



**HWA CHONG INSTITUTION**  
**JC2 Preliminary Examinations**  
**Higher 2**

CANDIDATE NAME

CT GROUP

20S7\_\_\_\_

CENTRE NUMBER

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INDEX NUMBER

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**BIOLOGY**

**9744/01**

Paper 1 Multiple Choice

**20 September 2021**

Additional Materials: Multiple Choice Answer Sheet

**1 hour**

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**INSTRUCTIONS TO CANDIDATES**

1. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.
2. Fill in your particulars on the Multiple Choice Answer Sheet. Write your **NRIC number** and shade accordingly.
3. There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.  
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.
4. At the end of the paper, you are to submit **only** the Answer Sheet.

**INFORMATION FOR CANDIDATES**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

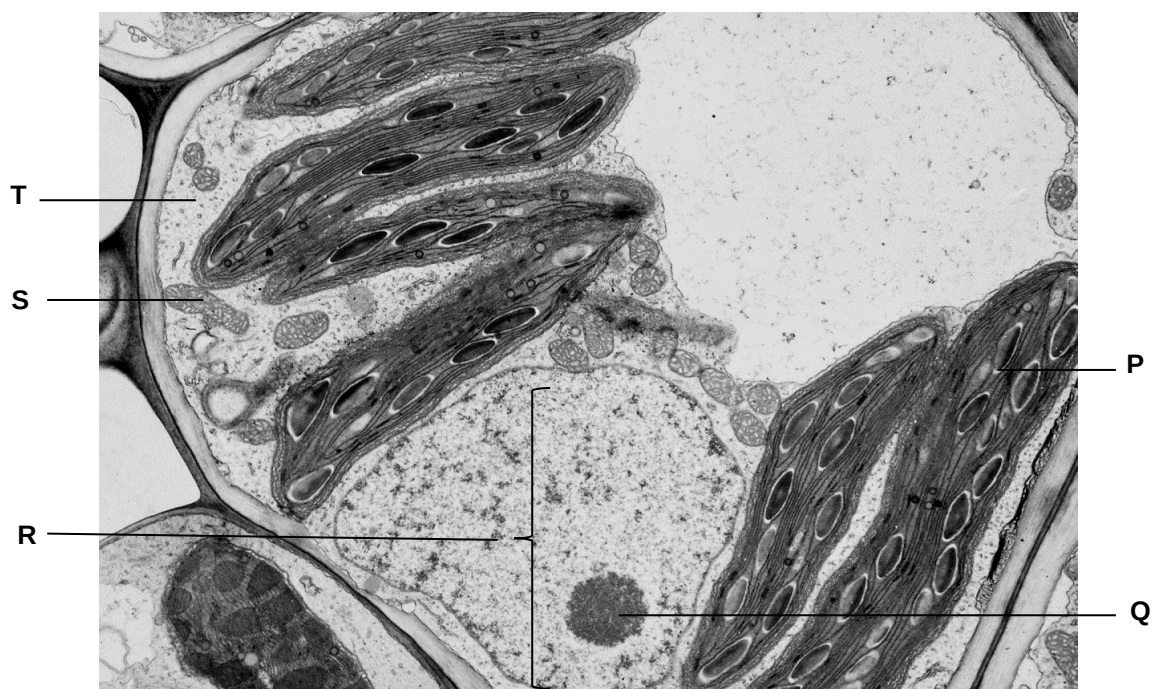
Any rough working should be done in this booklet.

The used of an approved scientific calculator is expected, where appropriate.

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This document consists of **22** printed pages.

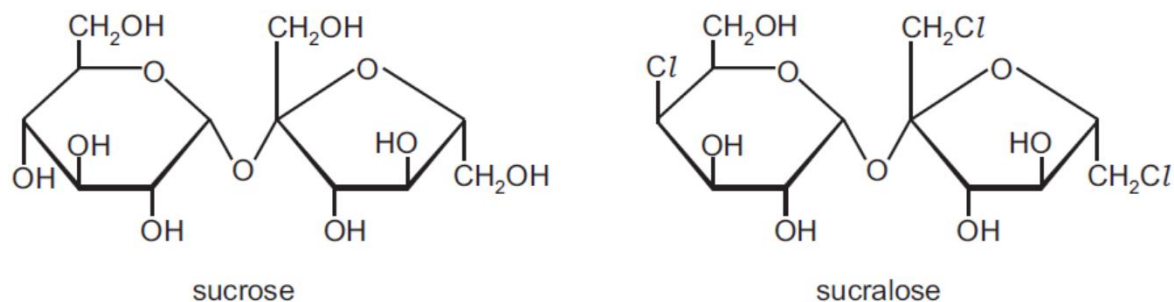
- 1 Which statement explains why viruses challenge the cell theory?
- A A HIV particle is 100 nm in size while a helper T cell is 5  $\mu\text{m}$ .
  - B Viruses contain only one type of nucleic acid.
  - C Viruses are able to change their genetic code to adapt to environmental changes.
  - D It is necessary for viruses to infect a host so as to complete its life cycle.
- 2 The electronmicrograph shows part of a plant cell with its components labelled **P** to **T**.



Which components produce ATP?

- A P, S and T
- B P, Q and R
- C Q, R and T
- D R, S and T

3 The diagram shows sucrose and sucralose.



The enzyme sucrase breaks down sucrose but not sucralose. Four students were asked to suggest why sucrase can break down sucrose but not sucralose.

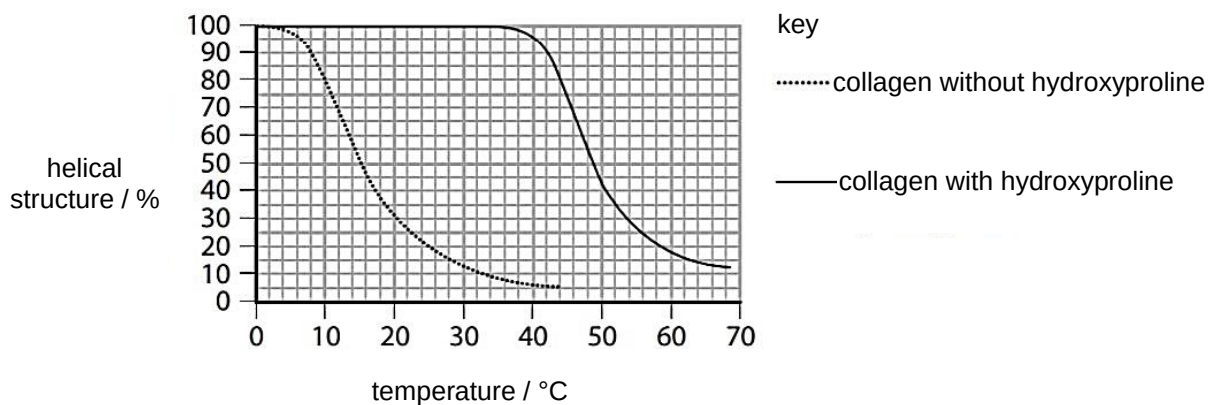
Three of the students gave correct suggestions.

Which suggestion **cannot** be correct?

- A The Cl atoms change the shape of the sucralose molecule so it is not the same shape as the active site of sucrase.
- B The Cl atoms of the modified fructose cannot bind to the active site of sucrase.
- C The Cl atoms cannot cause an induced fit after sucralose enters the active site of sucrase.
- D The Cl atoms cause fewer temporary hydrogen bonds between sucralose and the active site of sucrase.

4 Hydroxyproline and proline are components of collagen.

The graph shows the effect of temperature on the helical structure of collagen with and without hydroxyproline.



Which statements explain the importance of hydroxyproline in the structure of collagen?

- 1 Presence of hydroxyproline increases thermal stability of collagen.
- 2 At higher temperature, collagen with hydroxyproline results in less helices present than collagen without hydroxyproline.
- 3 Hydroxyproline contribute to the extensive hydrogen bonds that hold the helices together.

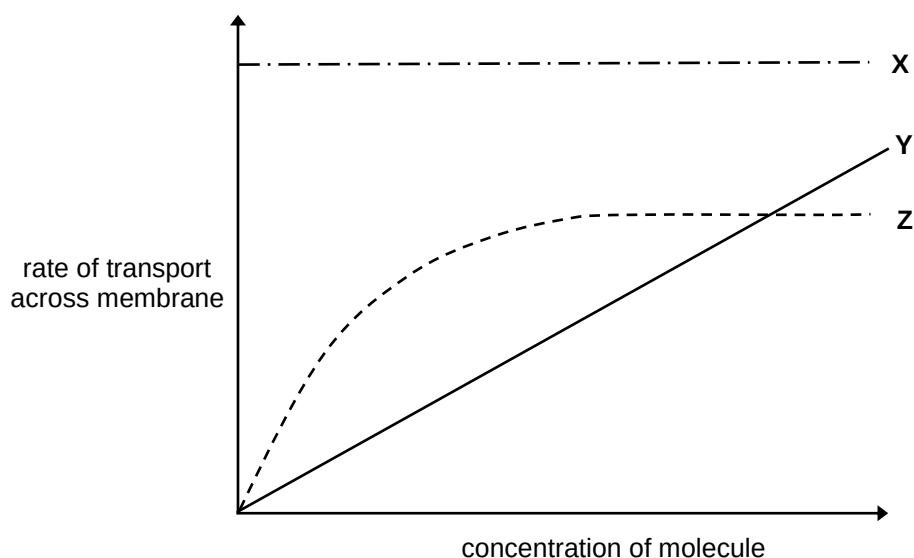
**A** 1 and 2

**B** 1 and 3

**C** 2 and 3

**D** 1, 2 and 3

- 5 The graph shows the effect of increasing concentration of molecules **X**, **Y** and **Z** on the rate of transport across the cell surface membrane into a cell.



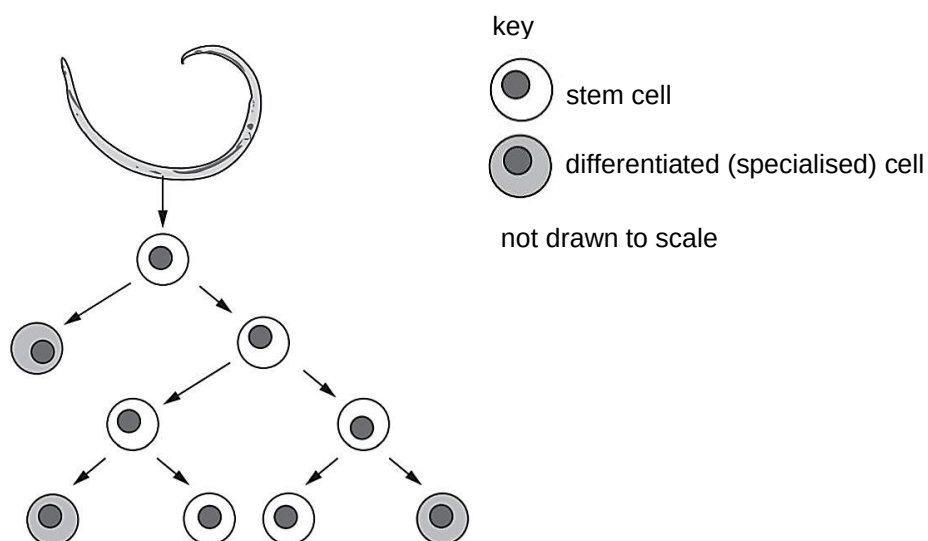
Which statements correctly describe how molecules **X**, **Y** and **Z** enter the cell?

- 1 **X** is a sodium ion that does not enter the cell passively.
- 2 **Y** is a small uncharged molecule that enters the cell by simple diffusion.
- 3 **Z** is a hydrophilic molecule that requires an additional protein to present for it to be transported across the membrane.

**A** 1 and 2 only      **B** 1 and 3 only      **C** 2 and 3 only      **D** 1, 2 and 3

6 Young nematodes have stem cells throughout the body.

The figure summarises three cell cycles of one of these stem cells.



Which statements correctly describe the role of stem cells in nematodes?

- 1 Stem cells continue to divide repeatedly to produce more stem cells.
- 2 At least three daughter cells retain the ability to divide.
- 3 Stem cells produce cells for growth, repair or replace worn out cells.
- 4 Stem cells differentiate to form tissues.

**A** 1, 3 and 4

**B** 1 and 2

**C** 2 and 3

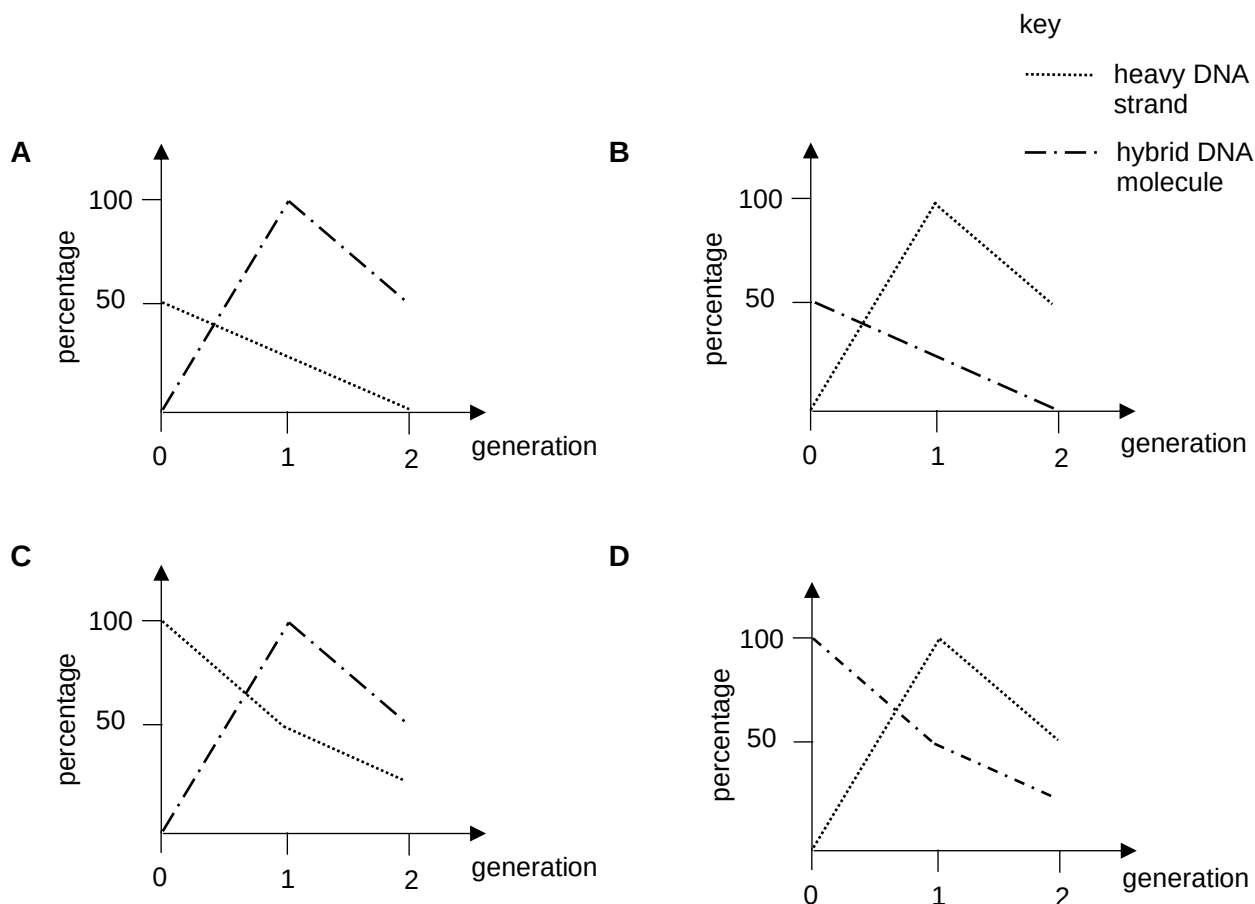
**D** 3 and 4

- 7 Meselson and Stahl investigated the mechanism of DNA replication in bacteria. They grew bacteria in a medium with only heavy nitrogen,  $^{15}\text{N}$ , until all of the bacterial DNA was fully labelled with heavy nitrogen.

These bacteria were moved from a heavy nitrogen medium and cultured in a medium with only light nitrogen,  $^{14}\text{N}$ , for two generations.

Hybrid DNA contains both heavy and light DNA.

Which graph shows the percentage of heavy DNA strands and the percentage of hybrid DNA molecules after two generations in the medium containing light nitrogen?



- 8 The following events occur during protein synthesis:

- synthesis of mRNA from DNA
- release of primary transcript from the DNA template
- elongation of growing polypeptide chain
- release of polypeptide from ribosome.

Which row is correct?

|          | number of events that involves bond breaking | number of events that involves bond formation |
|----------|----------------------------------------------|-----------------------------------------------|
| <b>A</b> | 2                                            | 2                                             |
| <b>B</b> | 3                                            | 2                                             |
| <b>C</b> | 4                                            | 3                                             |
| <b>D</b> | 4                                            | 4                                             |

9 Two students were discussing the involvement of DNA and RNA in transcription and translation.

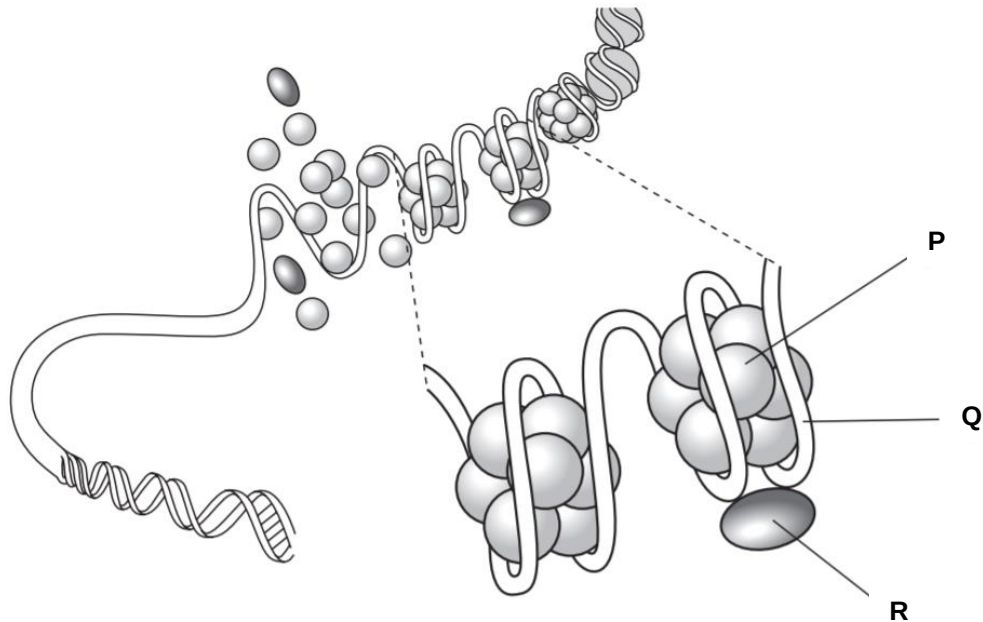
- Student 1 always stated correct facts.
- Student 2 gave further information, which was sometimes correct.

|   | correct facts given by student 1                                                    | further information given by student 2                                                      |
|---|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1 | A length of mRNA is 747 nucleotides long, including stop and start codons.          | This mRNA can produce a polypeptide that is 249 amino acids long.                           |
| 2 | Adjacent mRNA codons of AAU and CUG bind to complementary tRNA anticodons.          | There is a total of 14 hydrogen bonds formed between these two codons and their anticodons. |
| 3 | RNA polymerase catalyses the formation of the mRNA from the template strand of DNA. | During translation, an RNA adenine nucleotide will pair with a DNA thymine nucleotide.      |
| 4 | A DNA adenine nucleotide is structurally different to an RNA adenine nucleotide.    | The difference is in the hexose sugars; DNA is deoxyribose and RNA is ribose.               |

Which further information, given by student 2, is correct?

- A** 1, 2 and 3      **B** 1 and 3 only      **C** 2 only      **D** 2 and 4 only

- 10 The diagram shows part of the organisation of a DNA molecule and the associated histones.



Which statement about the features labelled **P**, **Q** and **R** during prophase of mitosis is correct?

- A** The DNA molecule **Q** wraps around the histone **R** to form small clusters that are held in place by histone **P**.
  - B** The DNA molecule **Q** and associated histone **P** move closer together as the chromosome condenses around **R**.
  - C** The DNA molecule **Q** wraps around histones **P** and **R** so that **Q** can fit inside the nucleus.
  - D** The DNA molecule **Q** and associated histones **P** and **R** form strands that fold and twist together to form a chromatid.
- 11 A single skin cell was isolated and transferred to growth medium in a sterile Petri dish. The Petri dish was incubated for 16 days. During this time, the number of skin cells multiplied as a result of repeated mitotic divisions.

One of the chromosomes in the nucleus of each skin cell has a telomere that contains many repeats of the base sequence TTAGGG. On each of days 4, 8, 12 and 16 of the incubation period, a single cell was removed from the Petri dish. The total number of bases in the telomere of this chromosome was determined for each cell. Each of the four cells had a different total number of telomere bases for this chromosome: 5548, 5580, 5645 and 5700.

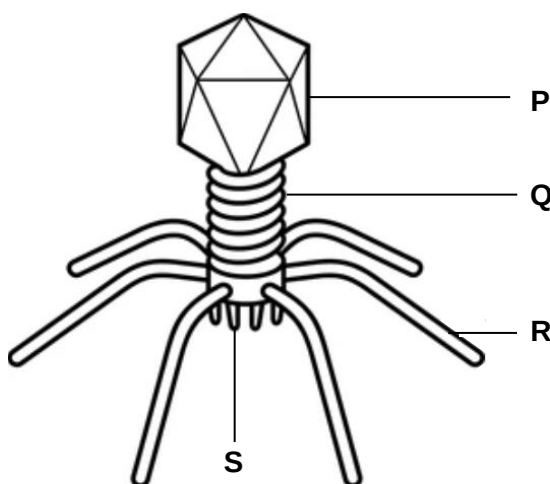
What was the total number of telomere bases in the chromosome from the cell that had undergone the most mitotic divisions?

- A** 5548
- B** 5580
- C** 5645
- D** 5700

12 Which statement is an example of post-transcriptional control of gene expression?

- A A gene product acting as a transcription factor for many other genes.
- B  $\beta$ -galactosidase activity in *Escherichia coli* exposed to lactose.
- C Liver cells beginning glycogenolysis in response to glucagon.
- D Different proteins produced from different mature mRNAs.

13 The diagram shows the structure of a T4 virus with its components labelled **P** to **S**.



Which statement is not correct?

- A Structure **P** is made up of multiple subunits and serve to contain double stranded RNA.
- B Structure **Q** is hollow and capable of contractile movement.
- C Structure **R** has a specific conformation that is complementary to the bacterial cell wall.
- D Structure **S** is where lysozyme is released to digest part of the bacterial cell wall.

14 Which statement about prokaryotic gene expression is correct?

- A The translation of prokaryotic mRNA can occur before transcription is completed.
- B A prokaryotic mRNA is considered polycistronic as the binding of many ribosomes on one mRNA strand allows for simultaneous translation.
- C The splicing of introns allows for a single mRNA gene to code for more than one protein.
- D The stop codon on the prokaryotic mRNA is found just before the polyadenylation signal sequence.

**15** Which statements correctly describe the transfer of DNA from one bacterium to another?

- 1 During conjugation, one strand of the F plasmid DNA is transferred to a new host via the mating bridge.
- 2 During transformation, crossing over must occur to result in an inheritable genetic change in the new host.
- 3 During transduction, the T4 bacteriophage transfers a DNA sequence adjacent to prophage insertion site into new host.

**A** 1, 2 and 3

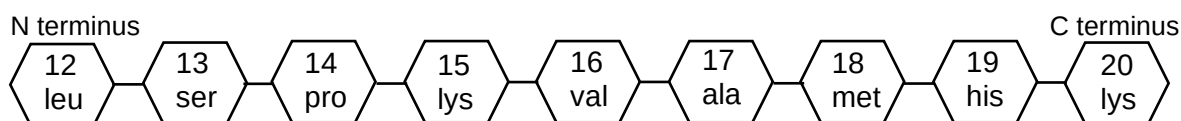
**B** 1 and 2 only

**C** 1 only

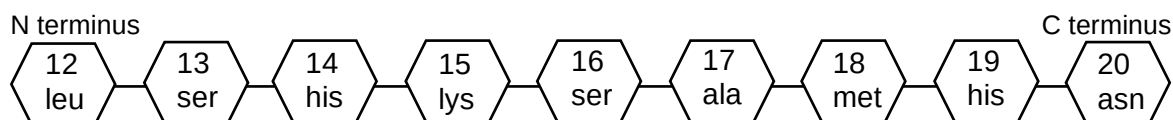
**D** 2 and 3 only

- 16** When a mutation occurs at the gene coding for protein A1, a structurally similar mutant protein, protein A2, is synthesised instead. The two proteins differ by only three amino acids, at positions 14, 16 and 20.

amino acid sequence of normal protein A1



amino acid sequence of mutant protein A2



The table shows the mRNA codons for the amino acids in these positions for proteins A1 and A2.

| amino acid | mRNA codons |
|------------|-------------|
| ala        | GCU         |
| ser        | AGU<br>UCA  |
| pro        | CCA         |
| leu        | CUU<br>UUA  |
| lys        | AAA         |
| ala        | GCU         |
| met        | AUG         |
| his        | CAU<br>CAC  |
| asn        | AAU         |
| val        | GUC         |

Which combination of mutations occurred in order to account for the amino acid sequence of the mutant protein A2?

- 1 Frameshift mutation resulting from an addition of a single nucleotide at position 14.
- 2 Substitution of a single nucleotide at position 14.
- 3 Frameshift mutation resulting from a deletion of two nucleotides at position 16.
- 4 Deletion of a single nucleotide at position 16.

**A** 1 and 3

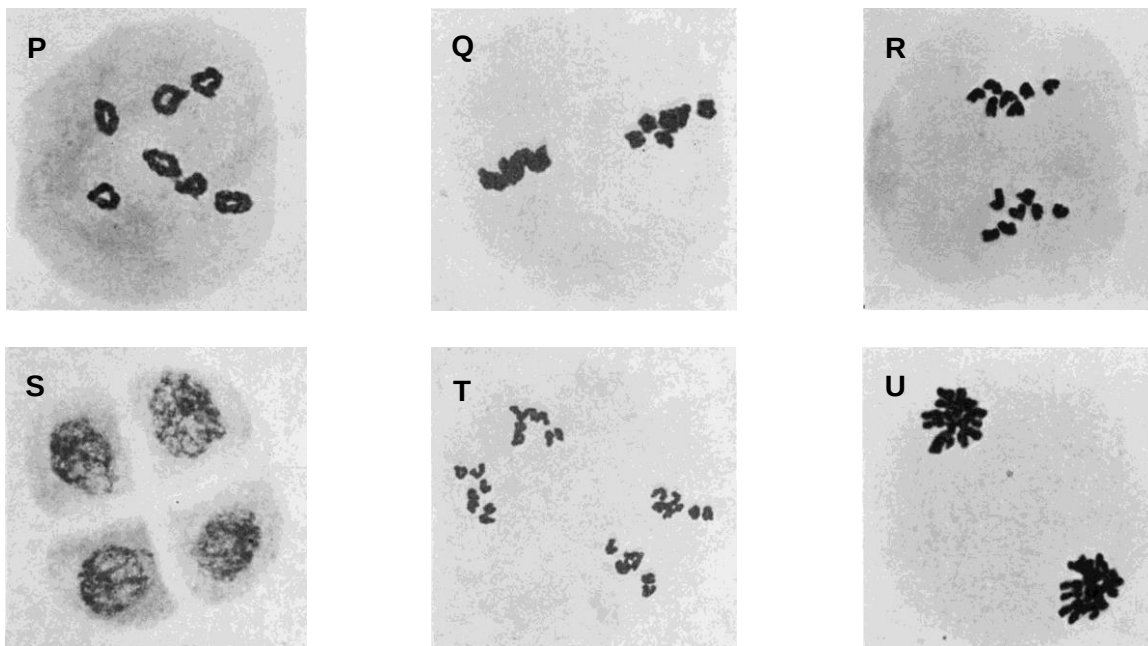
**B** 1 and 4

**C** 2 and 3

**D** 3 and 4

Refer to the following micrographs for **Q17** and **Q18**.

The micrographs show different stages of cell and nuclear division, labelled **P** to **U**, from one cell of a barley plant.



- 17** Which row correctly demonstrates the sequence of stages and the diploid number of chromosomes in the cell of the barley plant?

|          | sequence of stages                                                        | diploid number of chromosomes |
|----------|---------------------------------------------------------------------------|-------------------------------|
| <b>A</b> | $S \rightarrow P \rightarrow Q \rightarrow U \rightarrow R \rightarrow T$ | 14                            |
| <b>B</b> | $S \rightarrow P \rightarrow Q \rightarrow R \rightarrow U \rightarrow T$ | 7                             |
| <b>C</b> | $P \rightarrow R \rightarrow U \rightarrow Q \rightarrow T \rightarrow S$ | 14                            |
| <b>D</b> | $P \rightarrow Q \rightarrow R \rightarrow T \rightarrow U \rightarrow S$ | 28                            |

- 18** Which statement is correct?

- A** Paired homologous chromosomes are visible in micrographs **P** and **R**.
- B** If non-disjunction occurs in the cell shown in micrograph **Q**, the daughter cell will be a tetraploid.
- C** Cell wall and centrioles are forming in each cell in micrograph **S**.
- D** Only the nuclei, and not the cells, are dividing as shown in micrographs **T** and **U**.

- 19** Siamese cats have a temperature-sensitive tyrosinase that only functions in the cooler areas of the skin. This means they only produce a small quantity of melanin pigment. Melanin is mainly on their ears, face, paws and tail.

The Siamese allele of tyrosinase ( $t^s$ ) is recessive to the normal allele that causes full pigmentation all over the body ( $T$ ) but is dominant to the albino allele ( $t^a$ ).

A cross between an albino cat and a fully pigmented cat produced a fully pigmented cat. This fully pigmented offspring was then crossed with a Siamese cat. The genotype of the Siamese cat was unknown.

Which genotypes could be obtained for **all** possible offspring from the cross between the fully pigmented offspring and the Siamese cat?

- A**  $Tt^a$  and  $t^s t^s$
- B**  $Tt^s$  and  $t^s t^a$
- C**  $Tt^s$ ,  $t^s t^a$  and  $t^a t^a$
- D**  $Tt^s$ ,  $Tt^a$ ,  $t^s t^a$  and  $t^a t^a$

- 20** Normal eye colour in the fruit fly, *Drosophila melanogaster* is red but a white-eyed form exists. In the genetics of eye colour, red eye ( $R$ ) is dominant to white eye ( $r$ ) and the inheritance of eye colour is sex-linked (in a similar way to sex-linked conditions in humans).

The gene for body colour is located on an autosome. Normal body colour ( $B$ ) is dominant to ebony body colour ( $b$ ).

In a particular cross, a red-eyed female with normal body was crossed with a red-eyed male with normal body. The offspring produced are shown in the table.

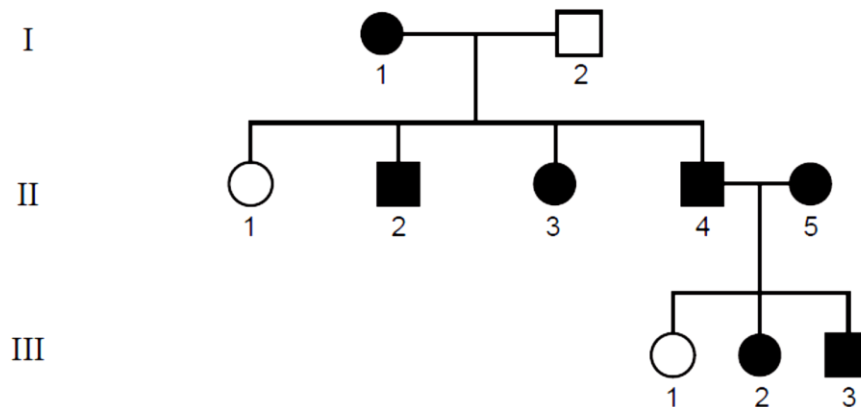
|         | red-eyed,<br>normal body | red-eyed,<br>ebony body | white-eyed,<br>normal body | white-eyed,<br>ebony body |
|---------|--------------------------|-------------------------|----------------------------|---------------------------|
| males   | 74                       | 26                      | 76                         | 24                        |
| females | 148                      | 52                      | 0                          | 0                         |

Which row correctly shows the genotypes of the parents?

|          | female parent | male parent |
|----------|---------------|-------------|
| <b>A</b> | $X^R X^R Bb$  | $X^R Y Bb$  |
| <b>B</b> | $X^R X^r Bb$  | $X^R Y BB$  |
| <b>C</b> | $X^R X^r Bb$  | $X^R Y Bb$  |
| <b>D</b> | $X^R X^r Bb$  | $X^r Y Bb$  |

**21** Familial atrial fibrillation is a condition in which individuals have an altered heart rhythm.

Shaded individuals in the pedigree have this condition.



Which statements are correct?

- 1 Individual I-1 must be homozygous since both her sons II-2 and II-4 have the condition.
- 2 Individual II-3 must be heterozygous as she would have inherited a normal recessive allele from her father I-2.
- 3 Individual III-1 must be homozygous as she would have received a dominant allele from each of her parents II-4 and II-5.

- A** 1 only  
**B** 2 only  
**C** 1 and 3 only  
**D** 2 and 3 only

- 22 The summer squash plant, *Cucurbita pepo*, produces edible fruits that vary in shape. The diagram shows the fruits of three different varieties of squash plants.



Patty pan  
(disc-shaped)



Di Nizza  
(spherical)



Alfresco  
(long)

Fruit shape in squashes is controlled by two genes, *A/a* and *B/b*, that are located on different chromosomes.

- A disc-shaped fruit is produced when both dominant alleles, *A* and *B*, are present.
- A spherical fruit is produced when either allele *A* or allele *B* is present, but not if both *A* and *B* are present.
- A long fruit is produced when both allele *A* and allele *B* are absent.

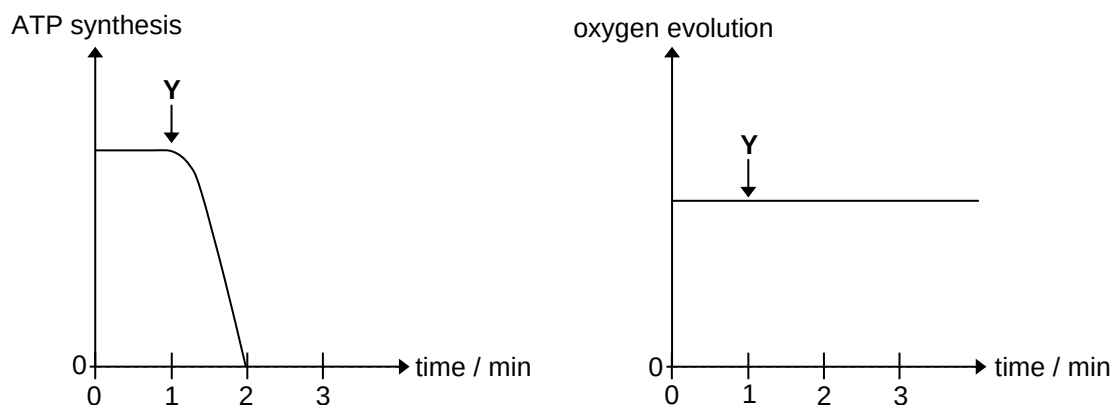
A gardener used pollen from a male flower of Alfresco to pollinate a female flower of Di Nizza. The gardener grew the seeds produced from this first cross and found that the offspring produced only spherical and long fruits.

The gardener also used pollen from a male flower of a double heterozygous Patty pan to pollinate a female flower of another double heterozygous Patty pan. He grew the seeds produced from this second cross and found that the offspring produced disc-shaped, spherical and long fruits.

Which row correctly represents the offspring phenotypic ratios for each cross?

|          | first cross          | second cross                          |
|----------|----------------------|---------------------------------------|
| <b>A</b> | 1 spherical : 1 long | 12 disc-shaped : 3 spherical : 1 long |
| <b>B</b> | 1 spherical : 1 long | 9 disc-shaped : 6 spherical : 1 long  |
| <b>C</b> | 3 spherical : 1 long | 12 disc-shaped : 3 spherical : 1 long |
| <b>D</b> | 3 spherical : 1 long | 9 disc-shaped : 6 spherical : 1 long  |

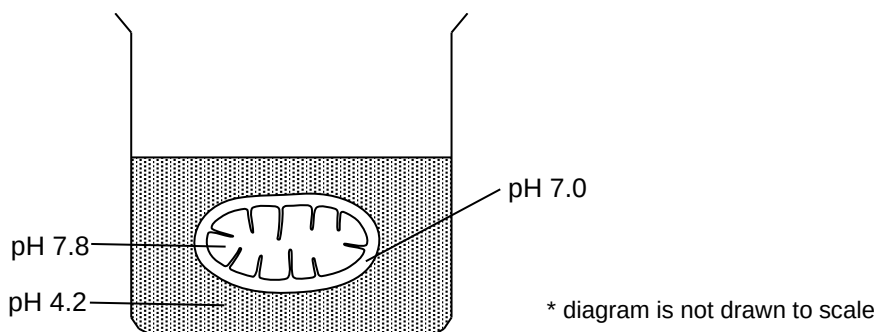
- 23** The effects of compound **Y** on chloroplast function have been determined by measuring the rates of ATP synthesis and oxygen production in suspensions of isolated chloroplasts placed in the presence of light. The results are shown with the arrow indicating the time of addition of compound **Y**.



Which statement for chloroplasts treated with compound **Y** is correct?

- A** No proton gradient exists across the thylakoid membrane and the reduction of NADP drops to zero.
- B** No proton gradient exists across the thylakoid membrane but the reduction of NADP continues at the initial rate.
- C** The proton gradient across the thylakoid membrane is unaffected but reduction of NADP drops to zero.
- D** No proton gradient exists across the thylakoid membrane but the reduction of NADP increases to a new, higher steady-state level.

- 24** An investigation was carried out to study the synthesis of ATP by oxidative phosphorylation. Mitochondria were isolated and placed in a low pH buffer in the presence of oxygen. This resulted in the synthesis of ATP.

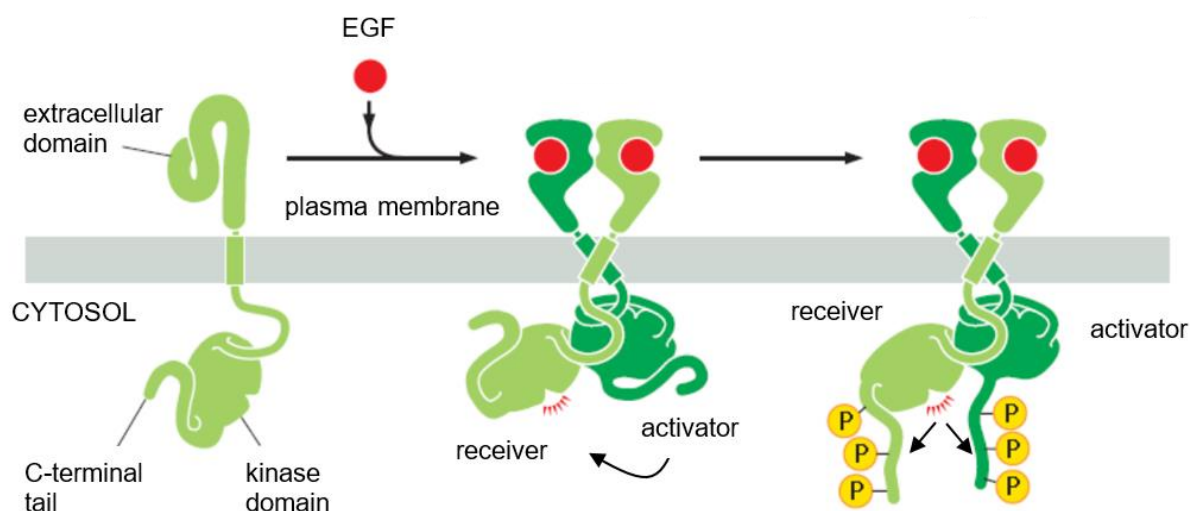


Which statement best explains the effect of low external pH on mitochondria?

- A**  $\text{OH}^-$  diffuses across the mitochondrial envelope and increases the  $\text{OH}^-$  concentration in the mitochondrial matrix so that its pH becomes 4.2.
- B** Diffusion of  $\text{OH}^-$  from the buffer into the inner membrane space increases.
- C**  $\text{H}^+$  diffuses across the mitochondrial envelope and increases the  $\text{H}^+$  concentration in the mitochondrial matrix so that its pH becomes 4.2.
- D** Diffusion of  $\text{H}^+$  from the intermembrane space into the mitochondrial matrix increases.

- 25 Many receptor tyrosine kinases (RTKs) are activated by auto-phosphorylation, but there are some exceptions.

The diagram shows the stages that occur to activate the RTK when epidermal growth factor (EGF) binds to EGF receptor.



Which row correctly describes the similarities and differences between the EGF receptor and an insulin receptor?

|          | similarity                                                         | difference                                                            |                                                                       |
|----------|--------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
|          |                                                                    | insulin receptor                                                      | EGF receptor                                                          |
| <b>A</b> | Both are activated by phosphorylation of the extracellular domain. | The dimer formed is symmetrical.                                      | The dimer formed is asymmetrical.                                     |
| <b>B</b> | Both are activated by phosphorylation of the intracellular domain. | The dimer formed is asymmetrical.                                     | The dimer formed is symmetrical.                                      |
| <b>C</b> | Both contain ligand-binding region.                                | The "receiver" kinase domain phosphorylates both receptor molecules.  | The kinase domain on each receptor molecule phosphorylates the other. |
| <b>D</b> | Both exist as inactive monomers before ligand binding.             | The kinase domain on each receptor molecule phosphorylates the other. | The "receiver" kinase domain phosphorylates both receptor molecules.  |

26 How many of these statements about second messengers are true?

- 1 All second messengers are water-soluble protein molecules.
- 2 Second messengers pass the signal on by binding to and altering the behaviour of selected signalling or effector proteins.
- 3 Second messengers are generated in large amounts in response to receptor activation and diffuse away from their source, spreading the signal to other parts of the cell.
- 4 Second messengers may allow for signal amplification to occur.

A 1

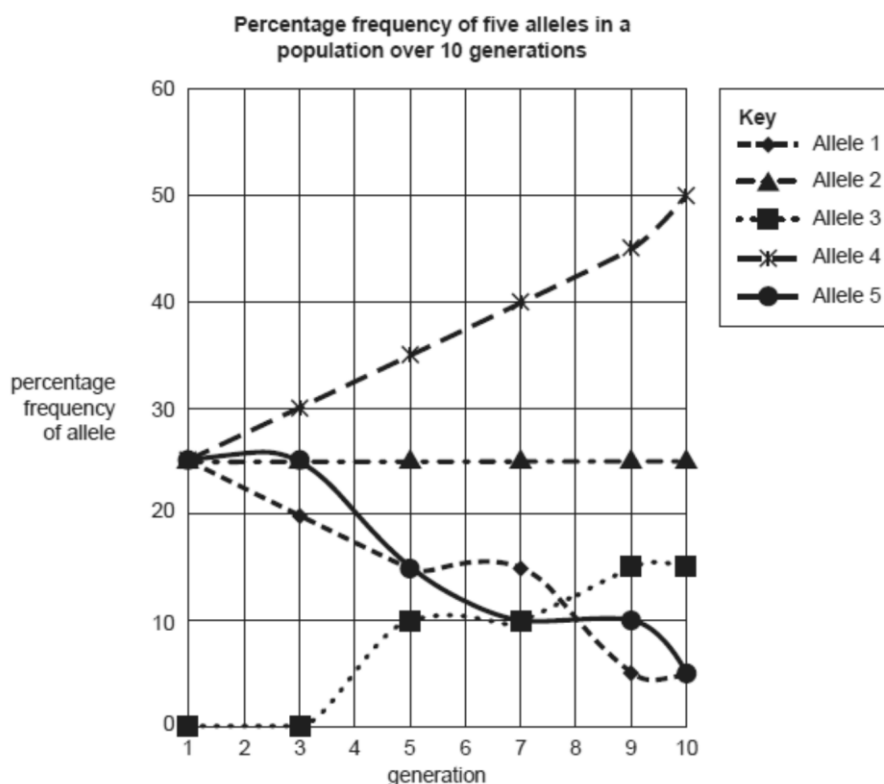
B 2

C 3

D 4

27 The graph shows the frequency of five alleles in a population over 10 generations.

The change in allele frequency is represented as a percentage of the frequency of the five alleles.



Based on the information in the graph, which statement is correct?

- A The founder effect can explain the change in allele frequency in the first three generations.
- B Genetic diversity within the population remains unchanged over the 10 generations.
- C The appearance of Allele 3 after generation 3 may be explained by gene flow.
- D A bottleneck event occurred between generation 7 and generation 9.

- 28** Members of two different species possess a similar-looking structure that they use in a similar fashion to perform the same function.

Which statement would best help distinguish between an explanation based on divergent evolution versus another on convergent evolution?

- A** The two species live at great distance from each other.
- B** The two species share many proteins in common and the nucleotide sequences that code for these proteins are almost identical.
- C** The sizes of the structures in adult members of both species are similar in size.
- D** Both species are well adapted to their particular environments.

- 29** Copies of a segment of DNA from species **X** were hybridised with copies of the homologous segment of DNA from species **P**.

The hybridised DNA segments were heated, and the temperature at which complete separation occurred (separation temperature) was recorded.

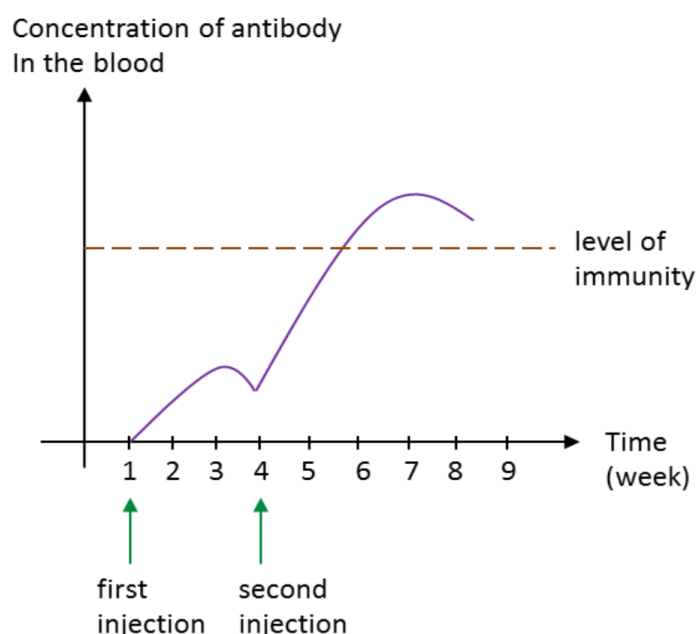
In order to determine evolutionary relationships, this process was repeated three times using DNA from species **Q**, **R**, and **S**. The results are shown in the table.

| hybridised DNA | separation temperature / °C |
|----------------|-----------------------------|
| <b>P-X</b>     | 30                          |
| <b>Q-X</b>     | 40                          |
| <b>R-X</b>     | 70                          |
| <b>S-X</b>     | 80                          |

Which statement is supported by the data in the table?

- A** Species **P** and species **X** share a more distant common ancestor as they have the least percentage difference in their DNA.
- B** Species **P** and species **Q** together have the same percentage difference in DNA from species **X** as has species **R**.
- C** Species **R** is more closely related to species **S** than it is to species **X**.
- D** Species **S** and species **X** share a more recent common ancestor as they have the least percentage difference in their DNA.

- 30 The graph shows the concentration of antibodies in the blood of a patient following two consecutive vaccine injections.



Which statement is a valid explanation for the results shown?

- A During the first injection, antigen-presenting cells undergo somatic recombination to phagocytose the pathogens and present antigens to helper T cells.
- B The adaptive immune response is only activated after the second injection and not after the first injection.
- C The secondary immune response is initiated at week 4 and results in clonal selection and expansion of memory B cells, mounting a larger immune response.
- D The adaptive immune response is initiated at week 4 and results in clonal selection and expansion of naïve B cells, leading to an increased antibody production.

---END OF PAPER---

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